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CHAPTER 9

Immobilization and Microbial Biosensors

Immobilization التثبيت

*when we immobilize something, the target will be the continuous use

*enzymes theoritically are recycled

- The technique of fixing the cells, organelles or proteins onto a solid support system, into a solid support matrix or retained by a membrane, in order to maintain stability and make possible their repeated or continued use.
- The early attempts of whole cell immobilization were developed from processes applied to the immobilization of enzymes and generally involved non-viable cells.

we still can immobolize them to produce something even if they are dead

- Immobilization is extended to viable cells as they were exploited in bioreactors and fermentation systems.

Immobilization of enzymes

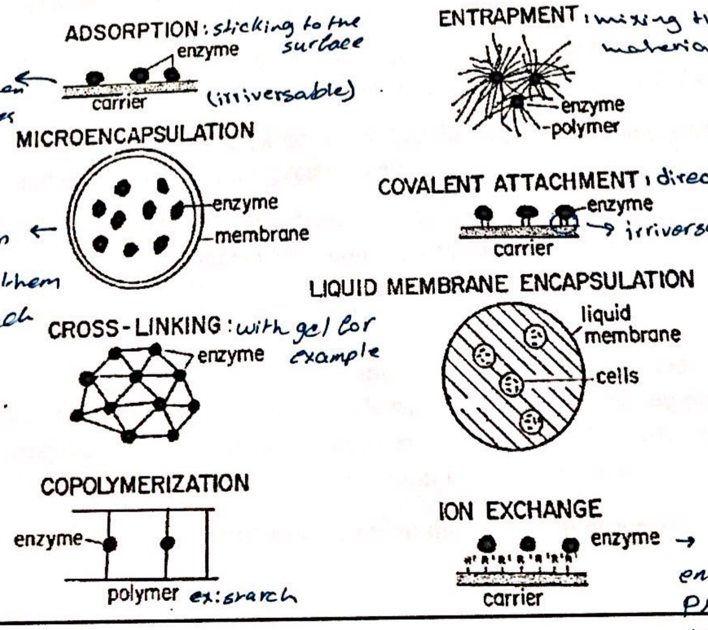
The advantages of immobilized enzymes beside reuse are as follows:

1. They can be easily separated from the reaction mixture containing any residual reactants and reused in subsequent conservations.
2. Immobilized enzymes are more stable over broad ranges of pH and temperature.
3. Enzymes are absent in the waste-stream
4. Immobilized systems specially lend themselves to continuous processes.
5. Reduced costs in industrial production.
6. Greater control of the catalytic effect.
7. Greater ease of new applications for industrial and medical purposes.
8. Immobilized enzymes permit the use of enzymes from organisms which would not normally be regarded as safe (i.e. non-GRAS).

* GRAS: generally regarded as safe

* higly pure enzymes: more expensive, don't need high amounts, high active units (IU/g)
* less pure enzymes: less expensive ~~crude~~, lower active units
* crude enzymes (not purified): cheapest, lowest active units

Strategies of enzyme immobilization



ADSORPTION: sticking to the surface (irriversable)

ex: it happens in nature when we add Abs to gold particles

ENTRAPMENT: mixing the enzyme with a random material

we microencapsulate certain things or drugs to protect them from the acidity of stomach so it won't be damaged

COVALENT ATTACHMENT: directional binding → irriversable bond

CROSS-LINKING: with gel for example

polymer ex: starch

ION EXCHANGE: enzyme → if I want to remove the enzyme we just change the PH which will change the ~~...~~ charge and it will be removed (reversable)

Immobilization of whole cells

Immobilized cells have the following advantages over conventional batch fermentation as well as over immobilized enzymes:

if I'm immobilizing a bacteria to produce an enzyme, I can regard the bacteria as a catalyst ~~because~~

1. When cells are immobilized the reactions are more homogeneous and can be treated more like catalysts.
2. The lag period which occurs in a conventional batch fermentation is eliminated for the accumulation of products associated with non-growth phase of the cells. if I immobolize bacteria, I can change the media easily without needing to transfer the bacteria
3. It is more feasible to run immobilized cells continuously at high dilution rates without the risk of washout which would occur in a conventional continuous culture system. why? because they are fixed
4. When cells are immobilized, they can be handled easily and thus can be easily transferred or simply decant the media and reuse. if there is no immobilization then I have to centrifuge to remove the bacteria which is not an easy process

Immobilization of whole cells

5. Higher and faster yield is possible because of the greater density of cells; furthermore, toxic materials are continuously removed. (while removing the media)
6. It is possible to recharge or resuscitate the cells by inducing growth and reproduction among resting cells.
7. A high capital cost is involved in installing, and operating a fermentor; in systems where comparison have been made, immobilized cells are cheaper than the conventional batch production.
8. The use of immobilized cells eliminates the need for enzyme extraction and purification. Furthermore, systems involving multi-enzyme reactions can occur more easily in intact cells harboring these enzymes and the immobilization keeps cells intact.
9. Immobilized cells are more suited to multiple step processes

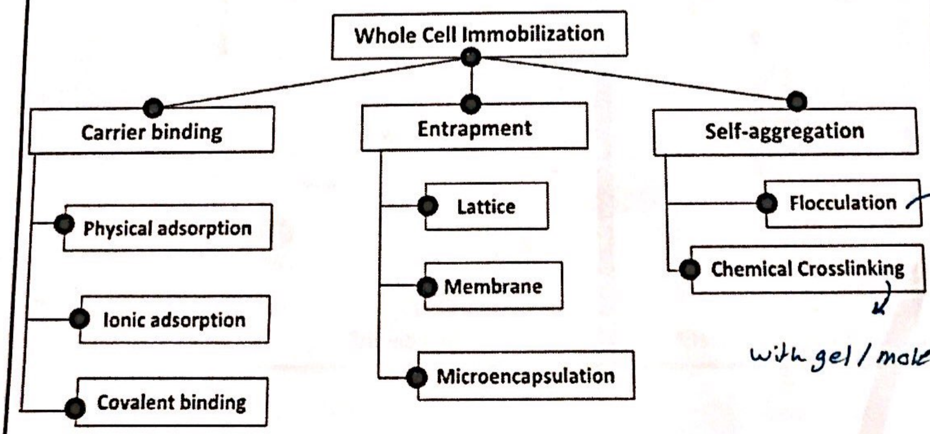
Immobilization of whole cells

Immobilized cells are particularly appropriate under the following conditions:

1. When the enzymes are intracellular: the use of immobilized cells would eliminate the need for breaking the cells for enzyme isolation.
2. When extracted enzymes are unstable during or after immobilization.
3. When the substrates and products are not high molecular weight substances.

* proteins are generally stable, but when they're removed from their place the become nonstable

Strategies of whole cell immobilization



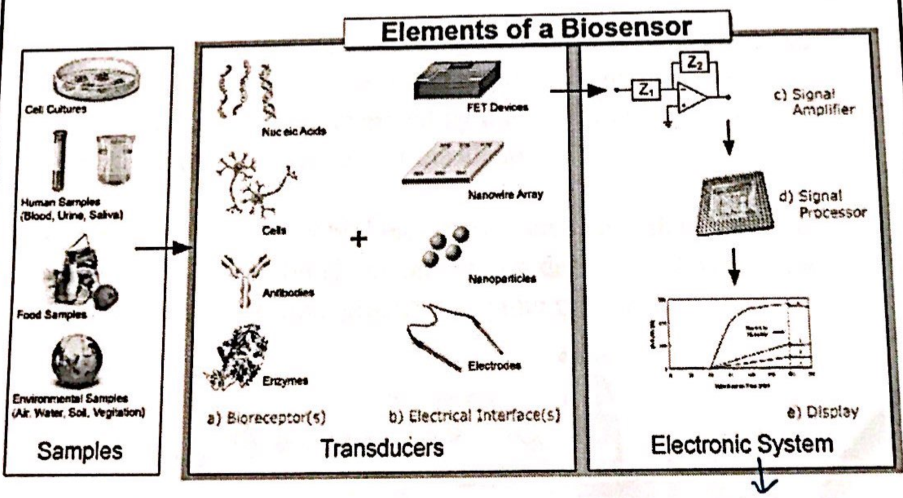
stacking up

with gel / molecules/ polymers, etc.

Biosensors ex. pregnancy test stick

- A biosensor is an analytical device which comprises of biological molecules as the recognition element with physical transducer and provides quantitative and semi-quantitative analytical data corresponding to the target concentration.
- Enzymes & whole cells have been exploited in developing biosensors, but the processes of isolation and purification of enzymes make them highly expensive which in turn enhance the cost of the sensor.
- Whole cells are a good alternative to enzymes as they are less expensive and have more stability. (enzymes may not be stable but the whole cell is stable)
- The basis of a microbial biosensor is the close contact between the microorganisms and the transducer which is generally based on immobilization on the transducer.

signal transducer → detector



detector

* biosensors : ① quantitative : exact mejorments (high / low / percentage)

② qualitative : yes / no

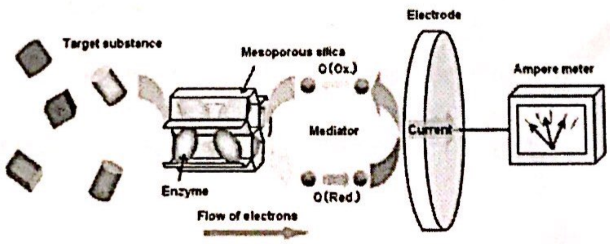
Microbial Electrochemical Biosensors

Microbial electrochemical biosensors are of three types:

1. Amperometric , chemical change
2. Potentiometric
3. Conductometric

Amperometric Biosensors

- Amperometric microbial biosensors work at a fixed potential with respect to the reference electrode and involves the detection of current generated by the oxidation or reduction of the species at the surface of the electrode.
- Amperometric microbial biosensors have been developed for the estimation of biochemical oxygen demand (BOD) for the measurement of biodegradable organic matter.



* oxidation and reduction will generate and electric current that's detected by the electrods

Biochemical oxygen demand (BOD) is the amount of dissolved oxygen needed (i.e. demanded) by aerobic biological organisms to break down organic material present in a given water sample at certain temperature over a specific time period.

The BOD value is most commonly expressed in milligrams of oxygen consumed per litre of sample during 5 days of incubation at 20 °C and is often used as a surrogate of the degree of organic pollution of water.

BOD reduction is used as a gauge of the effectiveness of wastewater treatment plants.

↳ used for mejorment

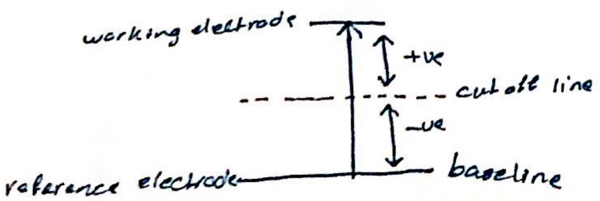
بحسب ال BOS قبل ال treatment وبعد ، وهاد بحدد مدى فعالية ال system اللي مستخدمينه للtreatment

* in waste water treatment, the more organic material = more oxygen demand = higher pollution

Potentiometric Biosensors

- The microbial potentiometric biosensor is developed using a conventional ion-selective electrode (pH, ammonium chloride, etc.) or a gas sensing electrode (pCO_2 and pNH_3) coated with an immobilized microbe layer. * the biosensor will detect the change in pH, ammonium chloride, etc.
- The microorganisms consume the analyte thereby bringing a change in the potential resulting from ion accumulation or depletion. * microbes will use a certain material (analyte, molecule)
- The potentiometric transducers compare the change between the working electrode and the reference electrode, and the signal is correlated to the concentration of the analyte.

* it's important to know the baseline for every single electrode



Potentiometric Biosensors

Table 12.8 Potentiometric microbial biosensors

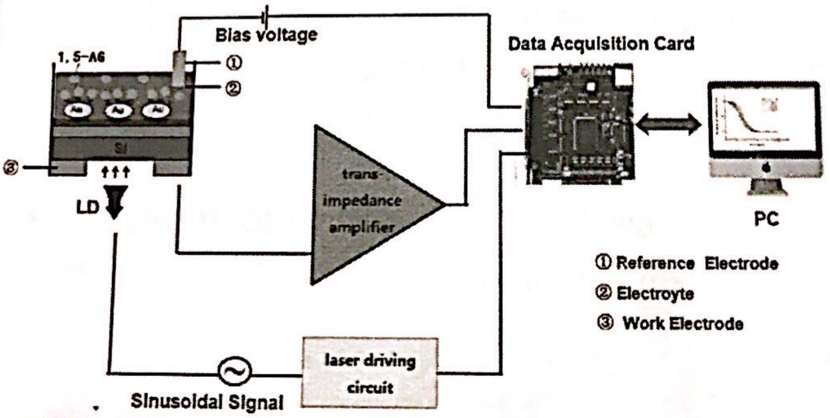
Microorganisms	Target	Limit of detection	Transducer
Pseudomonas aeruginosa	Cephalosporins	11 mM	pH electrode
Recombinant *E. coli*	Penicillins	5–30 mM	Flat pH electrode
Flavobacterium species	Organophosphates	0.025–0.4 mM	pH electrode
Recombinant *E. coli*	Organophosphates	3 µm	pH electrode
E. coli WP2	Tryptophan	0–12 µm	LAPS
Bacillus sp.	Urea	0.55–550 µm	NH_4^+ ion-selective electrode
S. ellipsoideus	Ethanol	0.02–50 µm	Oxygen
S. cerevisiae	Sucrose	3.2 µm	Oxygen

pesticides

*pesticides should have a withdrowl period before consuming the food it was sprayed on

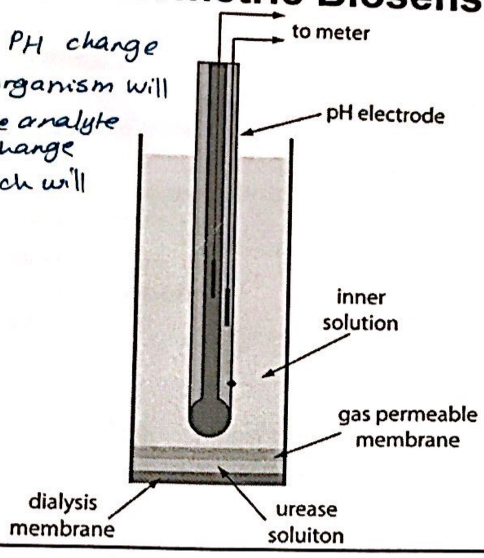
Light-addressable potentiometric sensor (LAPS)

*mejoring electric current



Potentiometric Biosensors

* mejoring PH change
* the microorganism will consume the analyte and it will change the PH which will be mejored



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Conductometric Biosensors

- Conductometry is a technique which measures conductivity changes in the solution due to the production or consumption of ionic species.
- A majority of microbe-catalyzed reactions involved a change in the ionic species.

* bacteria will produce ionic activity if it founde the analyte, if not, it will not produce ions, thus the analyte doesn't exist

Optical microbial biosensors

- Optical microbial biosensors are generally based on the changes in the optical properties brought by the interaction of the biocatalyst with the target analyte:
- Some of the optical properties are:
  1. Absorption (UV-Vis),
  2. Bioluminescence
  3. Chemiluminescence
  4. Reflectance
  5. Fluorescence
- The advantages of optical biosensors are compactness, flexibility, resistance to electrical noise and small probe size.

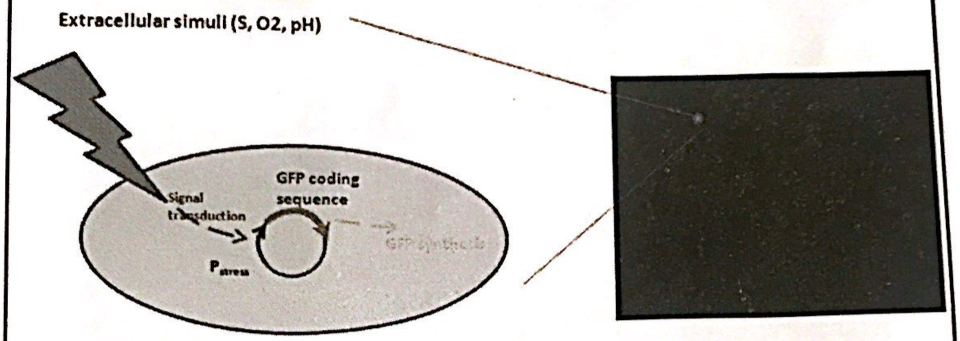
Optical microbial biosensors

Fluorescence Biosensor

- Fluorescence spectroscopy is a sensitive analytical technique which can detect very low concentrations of the analyte.
- There is a direct correlation between the fluorescence emission intensity and concentration at low analyte concentrations.
- The fluorescent microbial biosensors have been divided into two broad categories based on the detection mode: in vivo and in vitro.
- The in vivo fluorescent microbial sensor generally incorporates the use of a genetically engineered microorganism which is able to express a reporter gene encoding a protein.

Optical microbial biosensors
Fluorescence Biosensor

- Green fluorescent protein (GFP) encoded by the gene *gfp* is the most popular tool due to its stability and sensitivity



Optical microbial biosensors
Fluorescence Biosensor

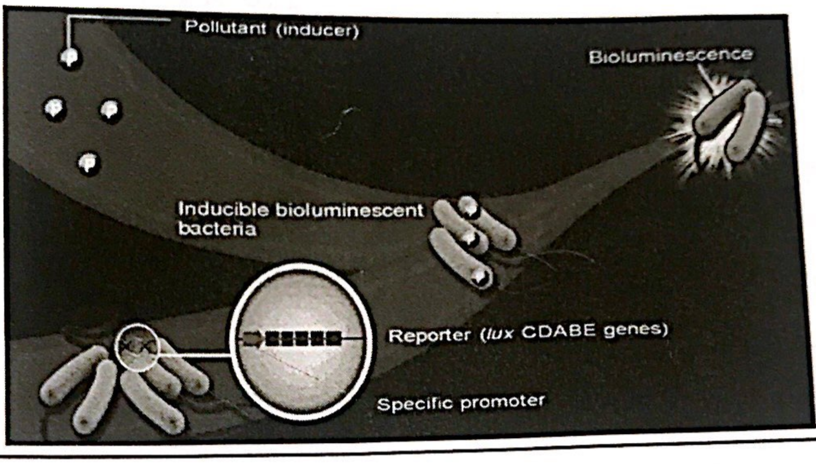
Table 12.9 Fluorescence microbial biosensors

Microorganisms	Target
E. coli ΔlysA mini-Tn5-Km-gfpmut3	Lysine
E. coli DH5α pTOLGFP	Toluene (Bioavailable BTEX)
Caulobacter crescentus NJH371	Uranium
Bacillus megaterium pSD202	Zinc
E. coli MG1655	Mitomycin C
Recombinant *Pseudomonas syringae*	Bioavailable iron
Pseudomonas fluorescence A506 (*pTolLHB*)	Bioavailable toluene

Optical microbial biosensors

Bioluminescent Microbial Biosensor

- The bioluminescent microbial biosensor measures the changes in luminosity of the living microorganisms



Optical microbial biosensors

Bioluminescent Microbial Biosensor

Table 12.10 Bioluminescence microbial biosenso

Microorganisms	Target
Ralstonia eutropha AE2515	Ni^{2+} and Co^{2+}
E. coli HMS174 with *mer-lux* plasmid pR27 of pRB28	Hg^{2+}
Recombinant E. coli (DL-2-haloacid dehalogenase and lux operon)	Halogenated organic acids
E. coli K12 pTetLux1	Tetracycline
E. coli K12 pTetLux1	Doxycycline
E. coli K12 pTetLux1	Chlortetracycline
V. fischeri	2,3,5,6,-tetrachlorophenol
E. coli EMS500	Ofloxacin
E. coli DH5α p TOLLUX	Toluene (Bioavailable BTEX)

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